MIL-I-23264A
3 July 1968
SUPERSEDING
MIL-I-23264
13 April 1962
(See 6.5)

MILITARY SPECIFICATION

INSULATORS, CERAMIC, ELECTRICAL AND ELECTRONIC, GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for electrical and electronic ceramic insulators fabricated from materials conforming to MIL-I-10. For purposes of this specification, the term "insulators" is used herein when referring to electrical and electronic ceramic insulators (see 6.1).

1.2 Classification. This specification shall cover only those insulators characterized by combining the specific designators of 1.2.1 to 1.2.5, inclusive.

NL	222x6/04T	B	56-	013
Class (1.2.1)	Insulating compound (1.2.2)	Finish (1.2.3)	Style (1.2.4)	Length (1.2.5)

1.2.1 Insulator designator. Insulators shall be identified by the two-letter designator NL.

1.2.2 Insulating compound designators. Insulators shall be further identified by appropriate designators of MIL-I-10, following the above designator, to represent insulating compound (see 6.2). The X-band and thermal conductivity designators of MIL-I-10 shall be applicable only when insulators are manufactured from insulating compounds designed for frequency applications from 100 MHz to 24 GHz or for high thermal conductivity applications.

1.2.3 Finish designator. Insulators shall be further identified by a single letter designator, B, W, or C, following the above designators, to represent finish (see 6.2).

Symbol	Finish 1/
B	Brown glaze. Glass and glass-bonded mica are not included.
W	White glaze. Glass and glass-bonded mica are not included.
C	Natural vitreous surface. Includes the fired ceramics, e.g., porcelain, steatite, and alumina; and the non-fired ceramics, e.g., glass and glass-bonded mica.

1/ Any special treatment or finish shall be as specified (see 6.2).

FSC 5970

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1.2.4 Style designator. Insulators shall be further identified by a three-character designator, a two-digit number followed by a hyphen, following the above designators, to represent style. Style numbers lower than 10 shall be preceded by a zero.

Style	Description
01 to 29, incl.	Standoff insulator
30 to 39, incl.	Feedthru insulator
40 to 49, incl.	Bushing insulator
50 to 59, incl.	Bowl insulator
60 to 75, incl.	Strain insulator
76 to 79, incl.	Spreader insulator
80 to 89, incl.	Pin insulator
90 to 99, incl.	Miscellaneous insulators, e.g., barrier, bead, cleat, disk, knob, etc.

1.2.5 Length designator. Insulators shall be further identified by a three-digit number, following the above designators, to represent the length of the part in units of 0.125 inch. The length designator numbers less than 100 shall be preceded by the appropriate number of zeros, i.e., 004, 010, etc.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-575 - Braid, Wire, (Copper, Tin-Coated Tubular).
QQ-B-621 - Brass Castings, Leaded Yellow.
QQ-B-626 - Brass, Leaded and Non-Leaded; Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strips).
QQ-B-637 - Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
QQ-B-639 - Brass, Naval: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-726 - Bronze Castings, Manganese and Alumium-Manganese.
QQ-B-750 - Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-576 - Copper Flat Products with Slit, Slit and Edge-Rolled, Sheared, Sawed or Machined Edges, (Plate, Bar, Sheet and Strip).
QQ-L-225 - Leaded Tin Bronze Castings and High Leaded Tin Bronze Castings.
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-624 - Steel Bars, Alloy, Cold Finished and Hot Rolled, (General Purpose).
QQ-S-633 - Steel Bars, Carbon, Cold Finished and Hot Rolled (General Purpose).
QQ-S-763 - Steel Bars, Shapes, and Forgings - Corrosion Resisting.
TT-E-527 - Enamel, Alkyd, Lustreless.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-676 - Boxes, Setup.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

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MILITARY

MIL-I-10 - Insulating Materials, Electrical, Ceramic, Class L.
MIL-P-116 - Preservation, Methods of.
MIL-S-890 - Steel: Forging and Bar for Hull, Engines, and Ordnance (Heat Treated).
MIL-F-1148 - Fibre, Vulcanized, Electrical and Mechanical Grades.
MIL-G-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness.
MIL-R-3065 - Rubber, Fabricated Parts.
MIL-S-15083 - Steel Castings (Navy).
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification (Navy).
MIL-C-45662 - Calibration System Requirements.

(See supplement 1 for list of applicable specification sheets.)

STANDARDS

FEDERAL

Fed. Std. No. 1 - Standard for Laboratory Atmospheric Conditions for Testing.
Fed. Std. No. 66 - Steel, Chemical Composition and Hardenability.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-456 - Electronic Parts, Date and Source Coding For.
MS15795 - Washer, Flat-Metal, Round, General Purpose.
MS33586 - Metal, Definition of Dissimilar.
MS35214 - Screw, Machine-Pan Head, Cross Recessed, Brass, Black Oxide, UNC-2A.
MS35273 - Screw, Machine-Drilled, Fillister Head, Slotted, Brass, Black Oxide Finish, UNC-2A.
MS35307 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Corrosion Resisting Steel, Passivated, UNC-2A.
MS35309 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Naval Brass, Plain Finish, UNC-2A.
MS35338 - Washer, Lock-Spring, Helical, Regular (Medium) Series.
MS35446 - Terminal, Lug, Solder Type, Copper Tubing, One Hole.
MS35690 - Nut, Plain, Hexagon, UNC-2B and UNF-2B.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheet.

3.2 Component material. When a definite material is specified (see 3.1), it shall be in accordance with the applicable specification or military standard (or requirement) listed in table I; however, when a definite material is not specified, a material shall be used which will enable the insulator to meet the performance requirements of this specification (see 3.1). Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

TABLE I. Component materials.

Component material	Requirement paragraph	Applicable specification or military standard
Ceramic parts	3.2.1	MIL-I-10
Metal parts:	3.2.2	
Brass	---	QQ-B-621, QQ-B-626, QQ-B-637, QQ-B-639
Bronze	---	QQ-B-726, QQ-B-750, QQ-L-225
Copper	---	QQ-B-575, QQ-C-576
Steel	---	QQ-S-624, QQ-S-633, QQ-S-763, MIL-S-890, MIL-S-15083, FED STD 66
Lockwashers	---	MS35338
Nuts	---	MS35690
Screws	---	MS35214, MS35273, MS35307, MS35309
Terminal lugs	---	MS35446
Washers	---	MS15795
Plating	3.2.2.2	QQ-N-290, QQ-P-416
Fibre	---	MIL-F-1148
Gaskets	---	MIL-G-1149, MIL-R-3065
Painting	---	TT-E-527, MIL-E-16400

3.2.1 Ceramic parts. Grade of material shall be as listed on the applicable specification sheet, or a numerically higher grade may be substituted, unless otherwise specified on the contract or order. The X-band designator must be as specified in the contract or order.

3.2.1.1 Glaze. Unless otherwise specified (see 1.2.3, 3.1, and 6.2), insulators covered by this specification shall be either brown glazed or white glazed; except for those which, as pressure molded, extruded, or otherwise processed, have a glass-like surface. The glaze shall be continuous and smooth.

3.2.1.2 Unglazed surfaces. Unless otherwise specified (see 1.2.3, 3.1, and 6.2), the unglazed surface or hang hole of glazed insulators shall be treated with a protective coating of a silicone compound (see 6.3) satisfactory to the Government. The silicone coatings shall be cured in accordance with the supplier's recommended techniques.

3.2.2 Metal parts. All current carrying metal parts shall be made of a nonferrous material, or as specified (see 3.1). Unless otherwise specified (see 3.1), roughness height shall not exceed 125 rms microinches for all exposed metal-bearing surfaces.

3.2.2.1 Dissimilar metals. Where dissimilar metals (as defined in MS33586) are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided.

3.2.2.2 Plating. All exposed corrodible ferrous parts shall be cadmium plated in accordance with class 1, 2, or 3, type II of QQ-P-416, except that a preliminary plating of other metal is permissible. When specified (see 3.1 and 6.2), other metal parts shall also be cadmium plated.

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3.3 Design and construction. Insulators shall be of the design, construction, and physical dimensions specified (see 3.1).

3.3.1 Screw threads. Screw threads for all bolts, machine screws, nuts, and threaded holes shall be of the national form in accordance with Handbook H28, except as modified in table II. Unless otherwise specified (see 6.2), screws smaller than size #6-32UNC shall not be used. Insulators shall have a minimum of six usable threads.

3.3.1.1 External. External threads shall accept a class 1 nut. The manufacturing limits shall be those of class 1 screws, coarse-thread and fine-thread series, in accordance with Handbook H28 with the following exceptions:

(a) The pitch diameter, lead, and thread angle shall be acceptable if the full length of the finished screw enters a gage representing a class 1 minimum nut.
(b) The maximum minor diameter shall be 0.005 inch less than the minimum minor diameter of a class 1 nut.
(c) The fit of screws in a class 1 maximum nut shall give a thread depth engagement at least equal to the minimum thread depth of a class 1 fit between a screw and nut. This is determined as follows:

$$\frac{(a - b)\ 100}{d} = \text{percent thread depth engagement}$$

Where:

a = outside diameter of thread
b = maximum minor diameter of a class 1 nut
d = double depth of thread.

3.3.1.2 Internal (threaded holes). Unless otherwise specified (see 3.1), internal threads shall accept a class 1 metal screw of specified thread number, size, and length of thread. Threaded holes shall be within the limits shown in table II for the size specified. All threaded holes shall be countersunk to a depth not exceeding 1/16 inch.

TABLE II. Threaded holes.

Size of threaded hole	Threads per inch	No go gage pin, diameter B (inch)
#6	32UNC	0.122
#8	32UNC	0.148
#10	1/ 32UNF	0.174
1/4	20UNC	0.222
5/16	18UNC	0.281
3/8	16UNC	0.337

1/ For style 81, the screw size shall be #10-24UNC.

3.3.2 Camber. The camber (see 6.4.5) shown in table III, shall apply to all surfaces of all shapes. Figure 1 shall be used to establish the ratios listed in table III. For flat shapes, disks, and plates, these limits shall apply only where the thickness is equal to or greater than the thickness determined by the given formulas. 2/ Where the thickness "a" 2/ is less than that determined, the permissible camber shall be as specified (see 6.2).

2/ For flat shapes, disks, and plates, the minimum thickness "a" (see 3.3.2) shall be computed as follows:
Where the ratio of the maximum superficial dimension to the largest dimension at right angles to it is less than 5, minimum thickness "a" in inches = $\sqrt{\frac{A}{10}}$. Where the same ratio is 5 or greater, minimum thickness "a" in inches = $\sqrt{\frac{A}{8}}$. In both cases "A" is the outline area in square inches.

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TABLE III. Camber.

$\frac{L}{a}$	Maximum camber 1/ (percent of L) to be applied to surfaces	
	Unglazed	Glazed
Less than 2	0.25	0.35
From 2 to less than 3	0.30	0.40
From 3 to less than 5	0.35	0.45
From 5 to less than 8	0.40	0.50
From 8 to less than 13	0.45	0.55
From 13 to less than 20	0.50	0.60
From 20 to less than 30	0.55	0.65
From 30 to less than 40	0.60	0.70
From 40 to less than 50	0.65	0.75

1/ In determining "maximum camber" permitted, values of "e" (see figure 1) less than 0.006 shall be disregarded.

3.3.3 Tolerances

3.3.3.1 Ceramic. Unless otherwise specified (see 6.2), the tolerances shall be as specified in table IV.

3.3.3.2 Metal. Unless otherwise specified (see 6.2), the tolerance for metal parts shall be ±.020 inch.

3.3.4 Ellipticity of tubes. The ellipticity shall be determined by dividing the maximum diameter by the minimum diameter measured in the same plane perpendicular to the axis. This quotient shall not exceed 1.02 and shall apply to cylindrical shapes having a wall thickness of at least 10 percent of the outside diameter. If the wall thickness at the thinnest section of the wall is less than 10 percent of the outside diameter, the same ratio shall not exceed 1.03.

3.3.5 Wall thickness and outside diameter of tubes. Unless otherwise specified, wall thickness shall be as follows: Minimum wall thickness of all cylindrical shapes shall not be less than 8 percent of the outside diameter, nor less than 1/32 inch, whichever is greater. Maximum wall thickness shall not be more than 20 percent of the outside diameter, nor more than 7/8 inch, whichever is smaller. The outside diameter of tubes shall not be less than 0.100 inch.

3.3.6 Parallel limits. Except for standoff insulators, rods, and tubes (see 3.3.6.1), parallelism of surfaces shall be considered satisfactory if the thickness measured at any point is within the established thickness tolerance of the piece.

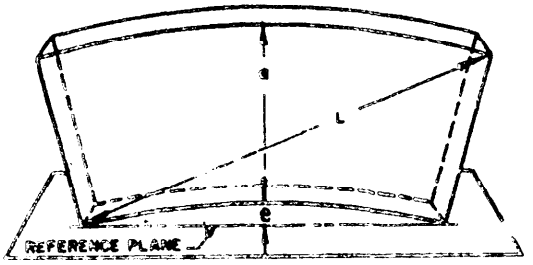
3.3.6.1 Standoff insulators, rods, and tubes. Parallelism of top and bottom surfaces shall be within 2 degrees, and these surfaces shall be perpendicular to the center axis within 2 degrees.

3.3.6.2 Bushing insulators. Both ends shall be either flat or concentrically concave within 0.010 inch.

3.3.6.3 Assembled insulators. The lower surface of the insulator base and the upper surface of the insulator cap shall be flat and parallel within 2 degrees of each other after assembly. These surfaces shall be perpendicular to the center axis of the insulator body within 2 degrees.

3.3.7 Alinement of holes. Holes shall be perpendicular to the axis or ends of the insulator, as applicable, and shall not vary by more than 2 degrees from this perpendicular.

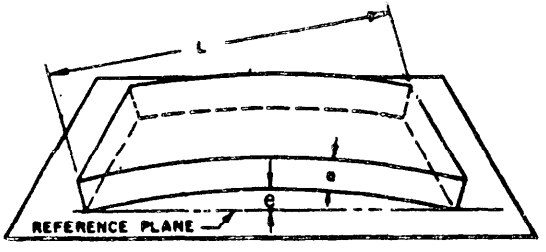
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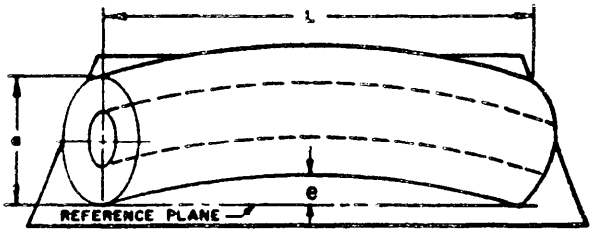
L = Maximum diagonal length on the face.
a = Minimum dimension in same direction as e.
e = Maximum perpendicular distance from chord to corresponding curved surface.



L = Maximum length of piece.
a = Minimum diameter or minimum dimension in same direction as e.
e = Maximum perpendicular distance from chord to corresponding curved surface.



L = Maximum diagonal chord length.
a = Minimum dimension in same direction as e.
e = Maximum perpendicular distance from chord to corresponding curved surface.



L = Maximum length of piece.
a = Minimum diameter or minimum dimension in same direction as e.
e = Maximum perpendicular distance from chord to corresponding curved surface.

FIGURE 1. Maximum camber.

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TABLE IV. Dimension tolerances.

Material	Type of dimensions	Dimensions up to and including 12-1/2	Dimensions over 12-1/2	Cylindrical shapes (Outside and inside diameters)	Hole centers[1]	Hole diameter[2]
Glass or glass-bonded mica[3]	Noncritical and critical	Inch ±0.010	Inch ±0.015	Inch ±0.015	Inch 12-1/2 dia and less, ±0.005; over 12-1/2 dia, ±0.007	Inch 0.500 dia and less, ±0.005; over 0.500 dia ±1%
	Wall	Not applicable	Not applicable	±0.005	Not applicable	Not applicable
Porcelain	Noncritical	±3% with ±3/16 max, ±1/32 min	±1.5%	Not applicable	Not applicable	Not applicable
	Critical Unglazed	±1.5% with ±0.015 min	±1.5%	±1.5% with ±0.015 min	±1.5% with ±0.015 min	±1.5% with ±0.010 min
	Thickness	±0.015	Not applicable	±0.015	Not applicable	Not applicable
	Glazed	±2% with ±0.1875 max	±1.5%	±1.5% + 0.015 with ±0.020 min	±1.5% with ±0.015 min	±1.5% with 0.020 min
Steatite and other fired ceramics (except porcelain)	Noncritical	±2% with ±1/8 max, ±1/64 min	±1%	Not applicable	Not applicable	Not applicable
	Critical Unglazed	±1% with ±0.005 min	±1%	±1%	Less than 0.500 dia ±0.005; 0.500 dia and greater ±1%	Less than 0.500 dia ±0.005; 0.500 dia and greater ±1%
	Thickness	±0.010	Not applicable	±0.010 (nominal wall)	Not applicable	Not applicable
	Glazed	±2% + 0.012	±2% + 0.012	1 inch dia or less, ±2% with ±0.012 min; over 1" dia, ±1% + 0.010	Less than 0.500 dia ±0.005; 0.500 dia and greater ±1%	Less than 0.500 dia ±0.015; 0.500 dia and greater ±3%
	Thickness	±1% + 0.012	Not applicable	±0.010 (nominal wall)	Not applicable	Not applicable

1/ If pin gages are used for inspecting hold-center spacings, the design of the gage should be such as to meet the specific requirements. If using pin gages, consideration shall be given to the tolerance between hole-centers as well as hole-diameter tolerance.

2/ Holes leading from the glazed surfaces shall have the tolerances for glazed surfaces applied except when specified that the glaze be removed from the hole, in which case the tolerances for unglazed surfaces shall apply. Holes which are not perfect circles shall have the same tolerances applied to the minor axis only.

3/ The flatness tolerance shall be ±0.0015 inch per linear inch. When angles or V cuts are required in the edges of flat pieces, the tolerance on such angular dimensions shall be ±1 degree.

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3.3.8 Edges. All edges, where practicable, shall be beveled or rounded to prevent chipping.

3.4 Thermal shock. When insulators are tested as specified in 4.6.2, there shall be no evidence of damage.

3.5 Cantilever (when applicable). When insulators are tested as specified in 4.6.3, they shall withstand the ultimate cantilever loads specified (see 3.1).

3.6 Tensile (when applicable). When insulators are tested as specified in 4.6.4, they shall withstand the working, average, and ultimate tensile loads specified (see 3.1).

3.7 Torque (when applicable). When insulators are tested as specified in 4.6.5, they shall withstand the ultimate torsional loads specified (see 3.1).

3.8 Compression (when applicable). When insulators are tested as specified in 4.6.6, they shall withstand the ultimate compressive loads specified (see 3.1).

3.9 Dielectric withstanding voltage (when applicable). When insulators are tested as specified in 4.6.7, they shall withstand the voltage specified (see 3.1), without evidence of breakdown or corona discharge.

3.10 Marking. Insulators shall be marked in accordance with MIL-STD-130. Unless otherwise specified (see 6.2), the type designation, and the manufacturer's code symbol in accordance with MIL-STD-456, shall be permanently and legibly marked in the approximate location specified (see 3.1). For insulators less than 1 inch in length and 1/2 inch in diameter, the marking shall consist of the manufacturer's code symbol followed by a hyphen and the last digit from the type designation part number (see 3.1 and appendix). For example:

81443-4 shall be used in lieu of NL222W01-004 81443
(Insulators marked as required by a superseded specification will be considered to have acceptable markings - see appendix.)

3.10.1 Unglazed. Insulators shall be marked by stamping or molding all necessary markings on the body of the insulator.

3.10.2 Glazed. Insulators shall be marked by stamping in or below the glaze, or by molding or etching as specified by the Government. All necessary markings shall be applied on the body of the insulator, unless otherwise specified (see 6.2).

3.10.3 Cross section. Items which are standardized only with respect to cross section and which are not required to carry individual part numbers under this specification, shall be marked with the component and grade symbols of the type designation (see 1.2.1).

3.10.4 Insulators having detachable hardware. Insulators having detachable hardware such as flanges, shields, or sleeves shall be identified by markings on each piece of detachable hardware in accordance with 3.10 and as specified (see 3.1). Stock hardware items such as nuts, bolts, screws, washers, studs, etc. need not be marked.

3.10.4.1 Insulator assemblies. Insulator assemblies having detachable hardware shall be marked in accordance with 3.10.4 and as specified (see 3.1).

3.11 Workmanship. Insulators shall be processed in such a manner as to be uniform in quality and shall be free from surface flaws such as chips, blisters, surface marks, and other defects that will affect life or serviceability. Rough or irregular edges caused by minute chips or nicks are not considered as impairing the life or serviceability of the insulator unless sharp smooth edges are specified (see 3.1).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the insulator, are in accordance with the applicable referenced specifications or requirements prior to such fabrication (see 3.3).

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at room ambient temperature, relative humidity, and pressure. The referee conditions shall be as specified in Federal Standard No. 1.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B.

4.5.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of completely assembled insulators of the same type designation, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations specified in table V.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level I. The acceptable quality level (AQL) shall be as specified in table V.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE V. Group A inspection.

Examination	Requirement paragraph	Method paragraph	AQL (in defects per 100 units)
Visual and dimensional - - - -	3.1, 3.2.1.1 to 3.3.8, incl, and 3.10 to 3.11, incl	4.6.1	1.5

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4.5.1.3 Group B inspection. Group B inspection shall be performed on sample units from lots which have been subjected to and passed group A inspection, and shall consist of the examinations and tests specified in table VI.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-3. The AQL shall be as specified in table VI.

TABLE VI. Group B inspection.

Examination and tests	Requirement paragraph	Method paragraph	AQL (in defects per 100 units)
Subgroup 1			
Dimensional	3.1, 3.3	4.6.1	4.0
Dielectric withstanding voltage (when applicable)	3.9	4.6.7	4.0
Subgroup 2			
Thermal shock	3.4	4.6.2	2.5
Cantilever (when applicable)	3.5	4.6.3	2.5
Tensile (when applicable)	3.6	4.6.4	2.5
Torque (when applicable)	3.7	4.6.5	2.5
Compression (when applicable)	3.8	4.6.6	2.5

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed subgroup 1 of group B inspection shall be delivered on the contract or order, if the lot is accepted. Sample units which have been subjected to subgroup 2 of group B inspection shall not be delivered on the contract or order.

4.5.2 Inspection of preparation for delivery. Sample packages or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 or the documents specified herein.

4.6 Methods of examination and test.

4.6.1 Visual and dimensional inspection. Insulators shall be examined to verify that the design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.2.1.1 to 3.3.8, incl, and 3.10 to 3.11, incl).

4.6.2 Thermal shock (see 3.4). Insulators shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

(a) Special mounting - The samples shall be mounted on a perforated rack that fits both the hot and cold chambers.
(b) Test-condition letter - C.
(c) Measurements before and after cycling - Not applicable.

4.6.3 Cantilever (when applicable) (see 3.5). The insulator shall be secured to the testing machine by the regular mounting bolts, or equivalent. Any universal testing machine of suitable capacity may be used provided the error in the loading range does not exceed 1 percent. The load shall be applied at right angles to the axis of the insulator at the upper surface of the cap.

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4.6.4 Tensile (when applicable) (see 3.6).

4.6.4.1 Standoff and feedthru insulators. Insulators shall be secured to the testing machine (see 4.6.3), by the regular mounting bolts, or equivalent. The load shall be applied to the cap or top of the insulator, as applicable, in line with the axis of the insulator.

4.6.4.2 Strain and spreader insulators. Insulators shall be secured to the testing machine (see 4.6.3), by connecting shackles to the ends of the insulator. The load shall be applied in line with the main axis of the insulator. Each shackle shall be clamped so as to prevent slipping of the shackle. Clamps shall be positioned so that the distance between the edge of the nearest clamp and the end of the insulator is a minimum of one insulator length. The diameter of the shackle pin used shall not exceed 85 percent of the diameter of the insulator hole.

4.6.5 Torque (when applicable) (see 3.7). Insulators shall be secured to the testing machine (see 4.6.3), by the regular mounting bolts, or equivalent. The load shall be applied at right angles to the axis of the insulator through a torque member so constructed that the insulator is not subjected to any cantilever stress.

4.6.6 Compression (when applicable) (see 3.8). The load (see 4.6.3), shall be applied between the lower cap and the base paralleled to the axis of the insulator.

4.6.7 Dielectric withstanding voltage (when applicable) (see 3.9). Insulators shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude of test voltage - The test voltage shall be as specified (see 3.1) and shall be maintained for a minimum of 60 seconds.
(b) Nature of potential - Ac.
(c) Points of application of the test voltage - As specified (see 3.1).

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Insulators shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Insulators shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Insulators shall be individually packaged in accordance with MIL-P-116, method III, insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. When the weight of an insulator exceeds 10 pounds, the unit container shall conform to PPP-B-636, class weather resistant, style optional, special requirement use. Closure and waterproofing requirements shall conform to the procedures specified for level A packing herein. In the event the weight limitations of PPP-B-636 are exceeded, boxes conforming to PPP-B-640, class 2, shall be used. The closure sealing, and banding requirements shall be as specified in the appendix thereto. Unit containers conforming to PPP-B-636 and PPP-B-640 shall serve as shipping containers.

5.1.1.5 Intermediate packaging. Insulators, packaged as described in 5.1.1.4, and whose packaged displacement is 20 cubic inches or less, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum tare and cube, and shall contain multiples of five unit packages, not to exceed 100 packages or ten pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 units.

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5.1.2 Level C. Each cleaned and dried insulator shall be individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The packaged insulators shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirement. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with waterproof tape, 2 inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636, using nonmetallic or tape banding only. Unit containers which serve as shipping containers require no further packing other than banding requirements.

5.2.2 Level B. Unless otherwise specified, the packaged insulators shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirement. Closures shall be in accordance with the appendix thereto. For Army, fiberboard container shall be class weather-resistant, as specified in level A.

5.2.3 Level C. The packaged insulators shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This pack shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package, intermediate, and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible.

5.5 Inspection. Inspection of military packaging shall be in accordance with 4.5.2.

6. NOTES

6.1 Intended use. Insulators covered by this specification are intended for use as electrical-insulating elements in components and parts, or as insulators in subassemblies and equipments. This specification should be used in the design of new equipment and, insofar as is practicable, replacement parts should also conform to these requirements. Many additional sizes and shapes of insulators not covered by this specification are now in use. Existing designs under which equipment is being and will continue to be fabricated are such as not to permit complete substitution with insulators conforming to the requirements of this specification. These requirements may not be suitable in all cases for direct replacement of existing insulators.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet, and the complete type designation (see 1.2.1 and 3.1).
(c) Special treatment, finish, or plating required (see 1.2.3, 3.2.1.1, 3.2.1.2, and 3.2.2.2).
(d) Screw thread size if other than that specified (see 3.3.1 to 3.3.1.2, incl).
(e) Permissible camber if other than that specified in table III (see 3.3.2).
(f) Ceramic tolerances and limits other than those specified in table IV (see 3.3.3.1).
(g) Metal tolerance if other than ±.020 inch (see 3.3.3.2).
(h) Marking of insulators if other than that specified (see 3.10 to 3.10.4.1, incl).
(i) Number of insulators in a unit package (see 5.1).

6.2.1 Indirect shipment. The preservation, packaging, packing, and marking specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

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6.3 Unglazed surfaces. Unglazed surfaces (see 3.2.1.2) may be treated using a protective silicone compound such as Dow Corning DC-200, General Electric SF-99, Union Carbide L-45, or equivalent.

6.4 Definitions. For the purpose of this specification, the following definitions shall apply:

6.4.1 Chips. Chips are depressions on the surface where a piece of material has been completely broken out.

6.4.2 Blisters. Blisters are bubbles, voids, bulges, or lumps greater than 1/16 inch in at least one dimension. Blisters may be on the surface or within the insulator body.

6.4.3 Holes.

6.4.3.1 Mounting. Mounting holes are those holes in the insulator which are used as points of support for the insulator or for other members of an assembly.

6.4.3.2 Other. Other holes are all holes, other than mounting holes, such as winding or top-off holes in coil forms, lead holes, perforations for ventilation or for weight reduction purposes, etc.

6.4.4 Surface marks. Surface marks are small discolored surface blemishes, color spots, and cavities. Discoloration formed by parting compounds used in the fabrication of unfired ceramic components shall be excluded from this definition.

6.4.5 Camber. Camber is defined in percent as the product $\frac{e}{L} \times 100$; where "e" is the maximum perpendicular distance from the chord to the corresponding curved surface and "L" is a reference length measured between the sector extremities (see 3.3.2 and figure 1).

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, INC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU
Navy - SH, EC
Air Force - 85
DSA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force -

Preparing activity:
Navy - EC

(Project 5970-0144)

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APPENDIX

CROSS INDEX OF TYPE DESIGNATIONS

10. SCOPE

10.1 This appendix cross indexes type designations described in this specification with those superseded in JAN-I-8, JAN-I-9, and JAN-I-21.

20. ORIGINAL JAN DESIGNATIONS

20.1 Type designations described in this specification compare with superseded types in JAN-I-8, JAN-I-9, and JAN-I-21 (see table VII).

TABLE VII. Cross index of type designations.

Specification	Type designation				
MIL-I-23264 (see 1.2.1)	NL Class	222 Insulating compound designators	B Finish	56- Style	013 Length
JAN-I-8 (see D-3)	NS Component	3 Grade	B Finish	46- Style	01 Part
JAN-I-9 (see D-3)	NA Component	3 Grade		46- Style	01 Part
JAN-I-21 (see D-3)	NP Component	2 Grade	B Finish	46- Style	01 Part
MS16229 1/ (see MS16229-1)	NS Component	L-422 Grade	B Finish	46- Style	01 Part

1/ For type designation IL-( )/U, see table VIII.

20.2 Class. Class designators NS, NA, and NP are replaced by "NL" (see 1.2.1).

20.3 Insulating compound designators. Original grade designators are replaced by three digits corresponding to the grades covered in MIL-I-10 (see 1.2.2). Unless otherwise specified (see 3.1), the first digit in the original grade designator remains the same with the second and third digits representing average dielectric strength and flexural strength requirements, respectively.

20.4 Finish. Finish designators B and W remain unchanged; C is substituted for U (see 1.2.3).

20.5 Style. Style designators are as shown in table VIII (see 1.2.4).

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TABLE VIII. Cross index.

MIL-I-23264		Style number			Standard MS16229	
Detail specification number	Style number	JAN-I-8	JAN-I-9	JAN-I-21	JECNS	Style number
MIL-I-23264/1	01	01	---	01	---	01
	02	02	---	02	---	02
	03	03	---	03	---	03
	04	04	---	04	---	---
	06	06	---	06	---	06
MIL-I-23264/2	10	10	---	10	---	---
	11	11	---	11	---	---
	12	12	---	12	---	---
	13	13	---	13	---	---
	14	14	---	14	IL51/U	14
	14	---	---	---	IL52/U	14
	14	---	---	---	IL53/U	14
	14	---	---	---	IL54/U	14
	14	---	---	---	IL55/U	14
MIL-I-23264/3	19	25	---	25	---	---
	20	20	---	20	---	20
MIL-I-23264/4	21	30	30	30	IL47/U	---
MIL-I-23264/5	26	---	---	39	---	---
MIL-I-23264/6	27	---	---	39	IL49/U	---
MIL-I-23264/7	28	---	---	38	IL50/U	---
MIL-I-23264/8	29	---	---	39	IL48/U	---
MIL-I-23264/9	30	---	---	---	IL19/U	---
MIL-I-23264/10	31	---	---	---	IL18/U	---
MIL-I-23264/11	32	---	---	---	---	---
MIL-I-23264/12	33	---	---	---	IL25/U	---
MIL-I-23264/13	34	---	---	---	IL42/U	---
	34	---	---	---	IL43/U	---
	34	---	---	---	IL44/U	---
MIL-I-23264/14	35	---	45	---	IL45/U	---
	35	---	---	---	IL46/U	---
	36	---	---	---	---	---
MIL-I-23264/15	37	---	---	48	IL20/U	---
MIL-I-23264/16	40	40	---	40	---	41
	41	41	---	41	---	42
	42	42	---	42	---	---
MIL-I-23264/17	43	---	---	---	IL37/U	---
	43	---	---	---	IL38/U	---
	43	---	---	---	IL39/U	---
	43	---	---	---	IL40/U	---
MIL-I-23264/18	55	45	---	45	---	---
	56	46	46	46	---	46
	57	---	47	47	---	---
MIL-I-23264/19	58	---	---	---	IL21/U	---
	58	---	---	---	IL22/U	---
	58	---	---	---	IL23/U	---
MIL-I-23264/20	64	50	50	50	---	---
	65	51	---	51	---	---
	66	52	---	52	---	---
	67	---	---	---	IL31/U	---

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TABLE VIII. Cross index - Continued.

MIL-I-23264		Style number			Standard MS16229	
Detail specification number	Style number	JAN-I-8	JAN-I-9	JAN-I-21	JECNS	Style number
MIL-I-23264/21	68	---	---	---	IL30/U	---
	69	---	---	---	IL32/U	---
	70	---	---	---	IL33/U	---
	70	---	---	---	IL56/U	---
	71	---	54	54	IL26/U	---
	71	---	---	---	IL27/U	---
	71	---	---	---	IL28/U	---
	71	---	---	---	IL29/U	---
MIL-I-23264/22	60	(Replaces MIL-I-17913)				
	61	---	---	---	---	---
	73	---	---	56	---	---
	74	---	---	57	---	---
	75	---	---	---	IL57/U	---
MIL-I-23264/23	76	55	---	55	IL34/U	---
	76	---	---	---	IL35/U	---
	76	---	---	---	IL36/U	---
MIL-I-23264/24	80	(Replaces MS91083)			IL41/U	---
	82	(Replaces MS91084)				
	83	(Replaces MS91085)				
MIL-I-23264/25	81	81	---	---	---	---

20.6 Part. The original two-digit part designator is replaced by a three-digit part designator (see 1.2.5). The last two digits remain the same as the original JAN designations, except where the original JAN numbers did not denote lengths in eighths of an inch.

30. SUPERSESSION DATA

30.1 This specification supersedes the following specifications in their entirety.

JAN-I-7, dated 31 May 1944, "Insulators, Glass-Bonded-Mica, Radio."
JAN-I-8, dated 29 April 1944, "Insulators, Steatite, Radio."
JAN-I-9, dated 31 May 1944, "Insulators, Glass, Radio."
JAN-I-21, dated 30 June 1944, "Insulators, Porcelain, Radio."
MIL-I-17918, dated 25 January 1961, "Insulator, Strain, Rigging."

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